Fiscal Year:	FY 2025	**Task Last Updated:**	FY 03/10/2025
PI Name:	Hyers, Robert Ph.D.		
Project Title:	Thermophysical Properties and Transport Phenomena Models and Experiments in Reduced Gravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	01609-2280	**Congressional District:**	2
Comments:	Formerly with University of Massachusetts at Amherst.		
Project Type:	Flight	**Solicitation / Funding Source:**	2015 NNH15ZTT002N MaterialsLab Open Science Campaigns for Experiments on the International Space Station
Start Date:	06/06/2017	**End Date:**	06/30/2024
No. of Post Docs:	1	**No. of PhD Degrees:**	1
No. of PhD Candidates:	1	**No. of Master' Degrees:**	1
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	3
No. of Bachelor's Candidates:	3	**Monitoring Center:**	NASA MSFC
Contact Monitor:	Phillips, Brandon	**Contact Phone:**	
Contact Email:	brandon.s.phillips@nasa.gov		
Flight Program:			
Flight Assignment:	NOTE: End date changed to 06/30/2024 per NSSC information (Ed., 3/8/24). NOTE: End date changed to 06/05/2023 per NSSC information (Ed., 3/8/23). NOTE: End date changed to 12/05/2022 per NSSC information (Ed., 6/6/22).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lee, Jonghyun Ph.D. (Iowa State University (formerly University of Massachusetts))		
Grant/Contract No.:	NNX17AL63G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The proposed program will consist of four major elements: continued development of novel non-contact methods for measuring thermophysical properties, including using electrostatic levitation in reduced gravity; application of these methods to materials of interest to industry and the international scientific community; application of the measured properties to produce accurate models of fluid flow and heat and mass transfer in the levitated samples; and application of the measured properties and models to test theories about the effect of processing on microstructure and material characteristics of a family of nonlinear optical crystals. The proposed elements primarily contribute to the research emphasis 1: Thermophysical Property Measurements, with supporting contributions to research emphasis; 2: Materials Processes Affecting Microstructure, Composition, and the Resultant Material Characteristics. The proposed elements vary in scope from the development of methods which are broadly applicable to measurement of thermophysical properties such as density and viscosity, to measurement of the properties of samples proposed by our team and other investigators in MaterialsLab, so models for specific experiments. Through the insight into the transport phenomena inside the samples, the experiment-specific models enable experiments that would otherwise be impossible. Similar models are currently in use for model-enabled experiments on the International Space Station. The results of the model-enabled experiments will advance the scientific understanding of the effect of processing on the properties of non-linear optical crystals, moving them closer to commercial production and sale here on Earth.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This project has the potential to improve the manufacturability of photonic devices based on photorefractive materials. Among the many applications are holographic storage of data. The ability to switch light with light could also have a large impact on telecommunications.
Task Progress:	• The Principal Investigator (PI) visited NASA Marshall Space Flight Center (MSFC) in August 2018 with graduate student Gwendolyn Bracker. We worked with MSFC staff to explore one method of fabricating samples for this investigation. We concluded that other methods should be explored for achieving the tight composition window needed for the proposed work. • After phasing discussions with MSFC, we agreed to push the work for Fiscal Year (FY) 2019 to the summer and fall, to allow continuity of funding for one or more students to prepare the bulk of the ground-based work in FY20. The funding profiles were adjusted accordingly. • PI and students visited MSFC Electrostatic Levitation Laboratory (ESL) Jul-Aug 2019. Trained students on principle and practice of ESL experiments. • Tested sample fabrication from oxide powders. Alternative approach was indicated for precise control of composition. • Surrogate material Bi¬2O3 was used to test processing in pressure ESL. • Commercial bismuth silicon oxide (BSO) and bismuth germanate (BGO) crystals were procured. • Characterization procedures were developed for these materials, using optical microscopy and x-ray diffraction. The characterized commercial crystals were confirmed to meet specifications. • PI and student Hodges visited MSFC ESL in January 2020. • Samples were fabricated from commercial crystals in platinum laser hearth. • Both BSO and BGO were processed successfully in ESL under pressure in N2/O2 gas mix. • Analysis of these measurements continued with remote work during COVID-19 “stay at home” orders. • Undergraduate Madeline Scott was honored with the University of Massachusetts Rising Researcher Award, April 2020, in part for her contributions to this project. This award serves to recognize the “most promising undergraduate students and to publicly acknowledge their excellent work”. Awarded to eight of almost 23,000 undergraduate students. • PI and grad student Beth Hodges participated in Professor Doug Matson’s International Space Station (ISS) operations on the Japanese Space Agency (JAXA) Electrostatic Levitation Furnace in August and November, 2020. • Results of MSFC ESL experiments were presented at the The Minerals, Metals and Materials Society (TMS) Annual Meeting, March 2022. • Impact testing at the University of Massachusetts confirms that samples processed in NASA MSFC ESL are strong enough for flight qualification. This was a critical concern for the flight experiments until the recent development. • NASA MSFC ESL experimental visit August 2022. This campaign has been delayed since the pandemic. It was determined that the full 5 atm gas pressure was needed to process the samples by ESL in 1 g. Returning to this capability required hardware changes to the MSFC ESL. • After discussions with ESL and other MSFC personnel, it was decided to delay the Science Concept Review until ESL experiments at NASA MSFC became possible. • Results to date were presented at the TMS Annual Meeting, Anaheim California, in March 2022 by MS student Elizabeth Hodges. • COVID significantly impacted collection of new data at MSFC ESL, analysis of prior results, and presentation of results at technical meetings. • Planned experimental campaign planned for late summer 2021 at NASA MSFC ESL was attempted in August 2022. Significant hardware upgrades were required. • The PI accepted a new position as the George I. Alden Professor and Head of the Department of Mechanical & Materials Engineering at the Worcester Polytechnic Institute, effective January 1, 2023. • Hardware upgrades for MSFC ESL were completed in the Spring of 2023. A new experimental campaign is planned if and when the grant is transferred/renewed.

Bibliography Type:	Description: (Last Updated: 03/30/2026)
Abstracts for Journals and Proceedings	Hodges E, Sansoucie M, Hyers R. "Containerless processing and characterization of potential host crystals for photorefractive devices." TMS 2022. 150st Annual Meeting and Exhibition, The Minerals, Metals and Materials Society, Anaheim, CA, February 27-March 3, 2022. Abstracts. TMS 2022. 150st Annual Meeting and Exhibition, The Minerals, Metals and Materials Society, Anaheim, CA, February 27-March 3, 2022. , Mar-2022